

ISTANBUL TECHNICAL UNIVERSITY Department of Electrical Engineering

ENGINEERING DRAWING (ELK 363E)

Orthographic Writing

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10/22/2023 1

TOPICS

- Views selection
- Alignment of views
- Orthographic writing steps
- Basic dimensioning
- Tangency and intersections

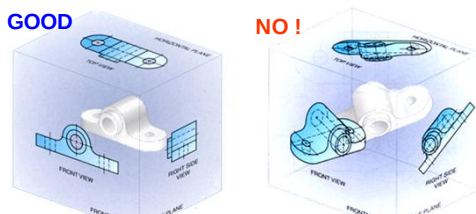
View Selection

VIEW SELECTION STEPS

1. Orient the object to the best position **relative** to a glass box.
2. Select the front view.
3. Select adjacent views.

STEP 1 : Orient the Object

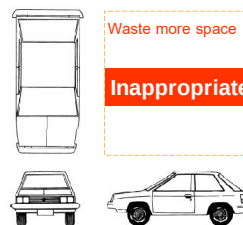
- The object should be placed in its **natural position**.
- The object should presents its features in **actual size** and **shape** in orthographic views.



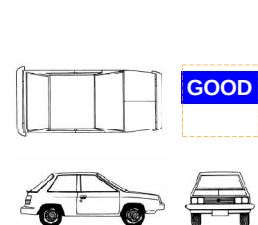
STEP 2 : Select a Front View

- The object's **longest dimension** should be presented as a **width**.

First choice



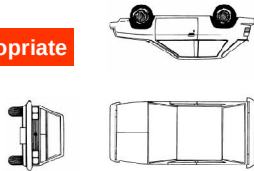
Second choice



STEP 2 : Select a Front View

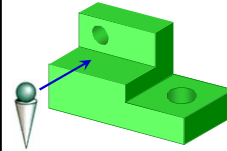
- The adjacent views that are projected from the selected front view should appear in its **natural position**.

Inappropriate

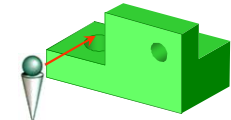


STEP 2 : Select a Front View

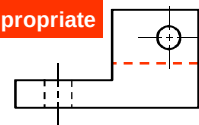
- Choose the view that have the **fewest number of hidden lines**.



GOOD

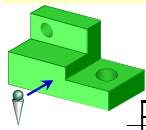


Inappropriate

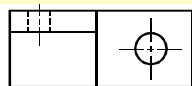


STEP 3 : Select an Adjacent View

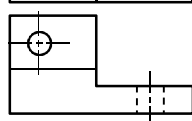
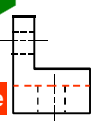
- Choose the view that have the **fewest number of hidden lines**.



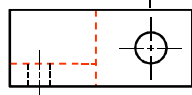
GOOD



Inappropriate



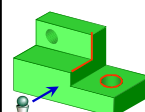
Inappropriate



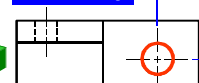
GOOD

STEP 3 : Select an Adjacent View

- Choose the **minimum** number of views that can represent the major features of the object.



Necessary



Hole's location can be specified on the same view.

Easy to understand

Difficult to interpret.



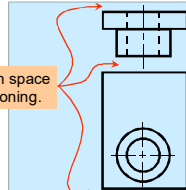
Necessary

STEP 3 : Select an Adjacent View

- Choose the views that are suitable to a drawing space.



Not enough space for dimensioning.



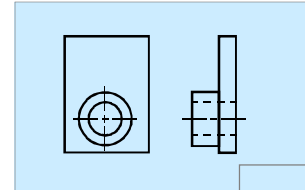
POOR

STEP 3 : Select an Adjacent View

- Choose the views that are suitable to a drawing space.



GOOD



Example : View selection

mislead to...

	F.V. & T.V.	Three views	F.V. & R.S.V.
Size description	✓	✓	✓
Shape description	✗	✓	✓

ONE-VIEW DRAWING

■ *Flat part* having a uniform thickness.

1 Thick

Unnecessary These 2 views provide only information about the part thickness !

ONE-VIEW DRAWING

■ *Cylindrical-shaped part.*

Unnecessary

Repeat !

Infer from CL

Unnecessary

TWO-VIEW DRAWING

■ There exists an identical view.

Repeat !

Unnecessary

TWO-VIEW DRAWING

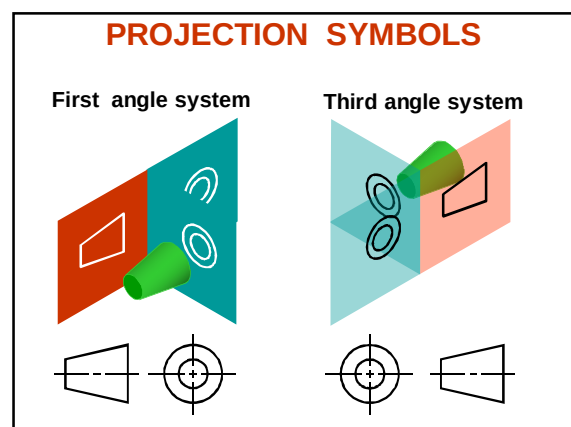
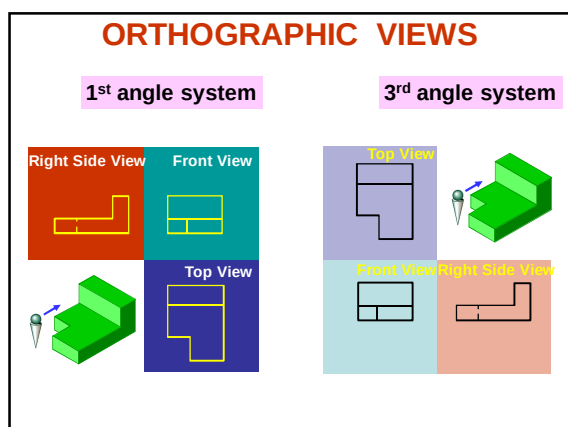
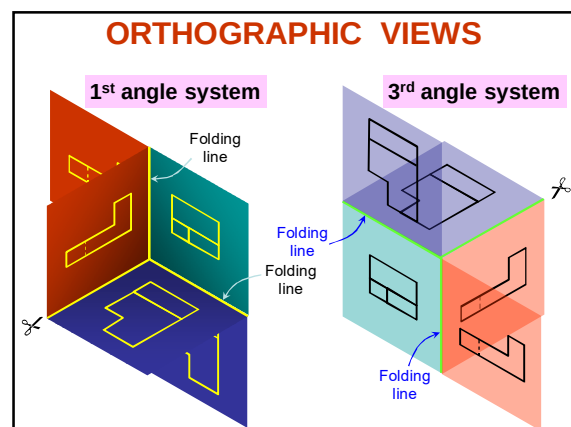
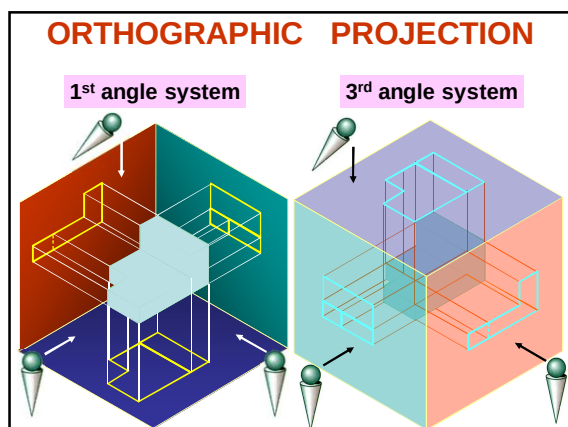
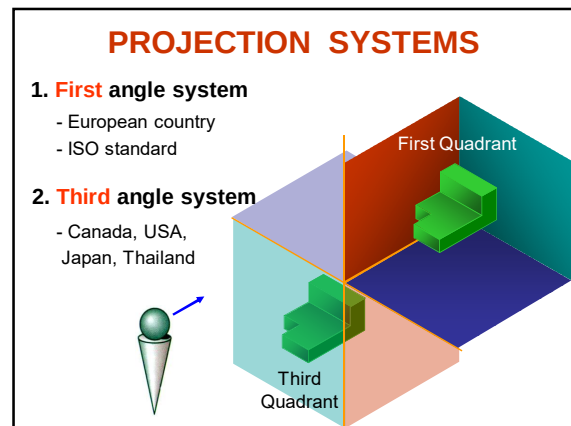
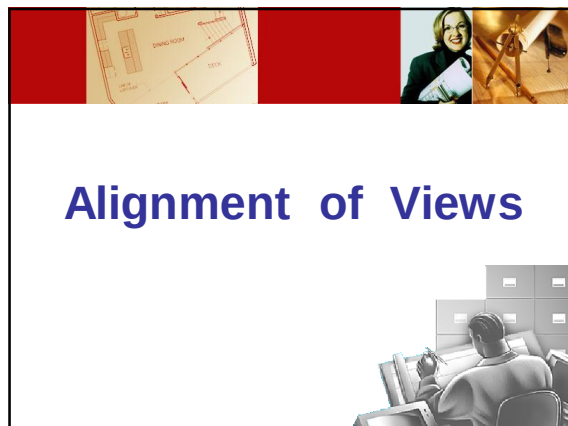
■ The 3rd view has no significant contours of the object.

Unnecessary

TWO-VIEW DRAWING

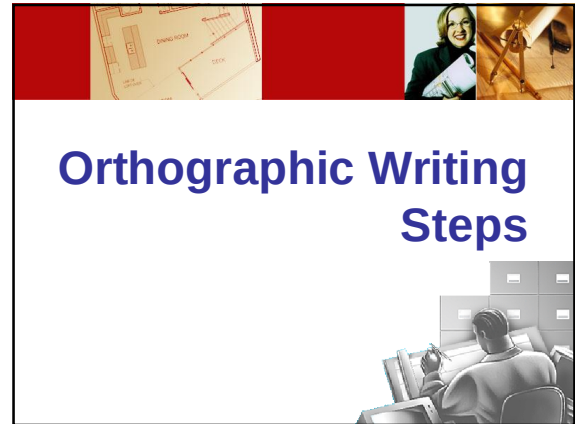
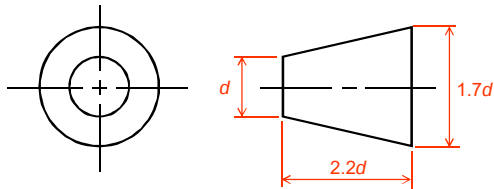
■ The 3rd view has no significant contours of the object.

Unnecessary



PROJECTION SYMBOLS

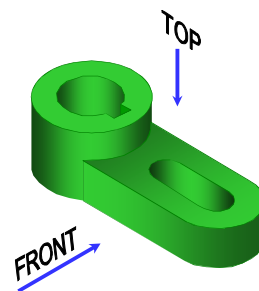
Suggested proportion



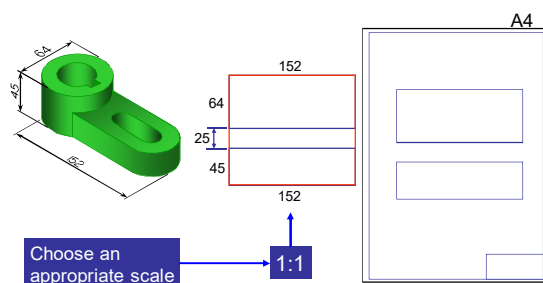
WRITING STEPS

1. Select the necessary views
2. Layout the views.
3. Project the views.
4. Dimension the views.

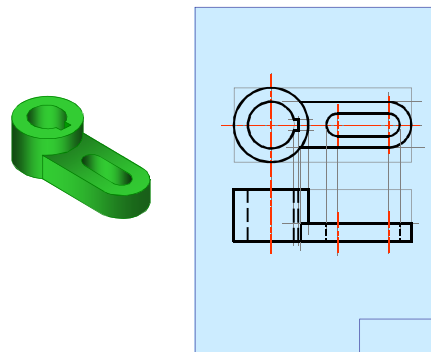
1. SELECT THE NECESSARY VIEWS

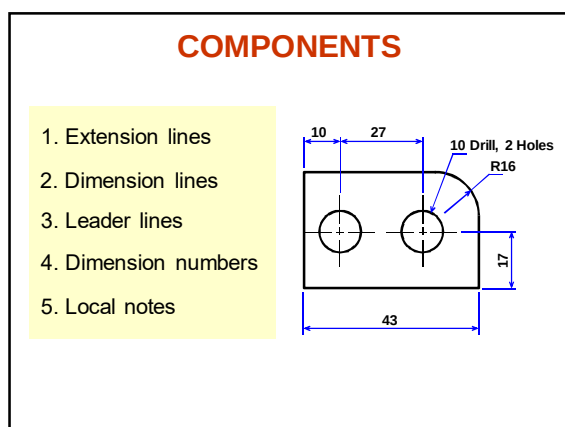
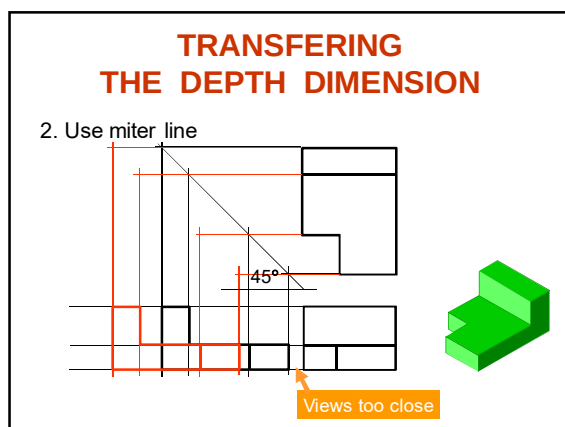
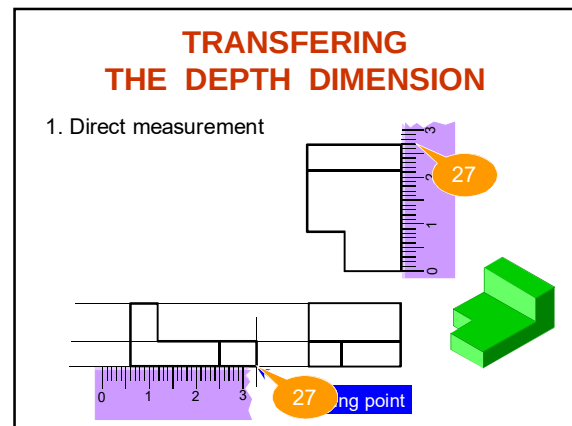
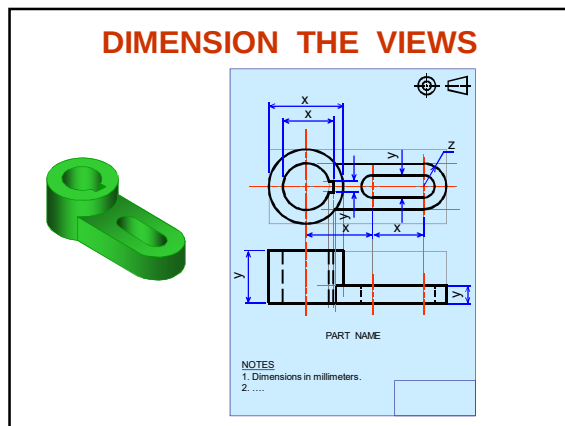


2. LAYOUT THE VIEWS



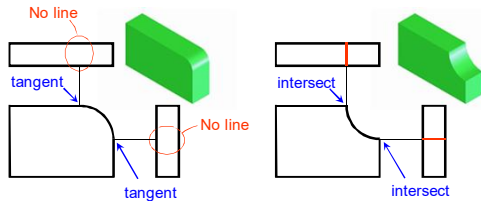
PROJECT THE VIEWS



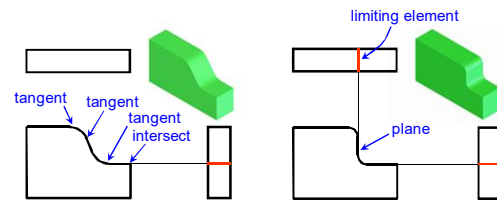


TANGENT & INTERSECTION

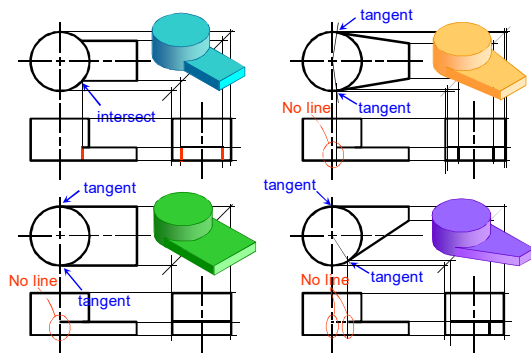
- No line is formed when curved surface **tangent** to a plane surface.
- Line is formed when curved surface **intersects** a plane surface.



TANGENT & INTERSECTION

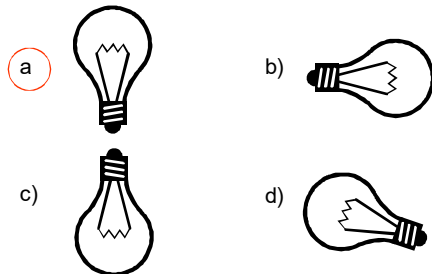


TANGENT & INTERSECTION



QUIZ

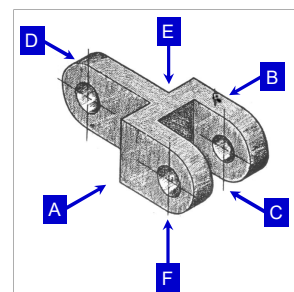
1. Which should be the **natural position** of the light bulb ? (20 sec)



0 5 10 15 20

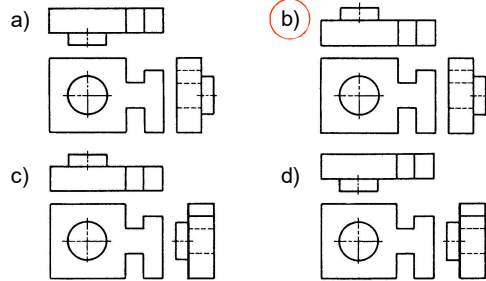
2. Which are the **necessary views** ? (60 sec)

- a) A-C-E
- b) E-B-D
- ☒ c) E-A
- d) E-C



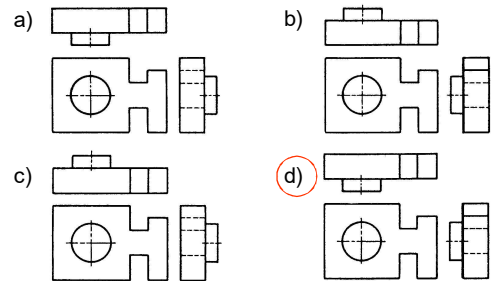
0 15 30 45 60

3. Which is in correct **first** angle projection ? (180 sec)



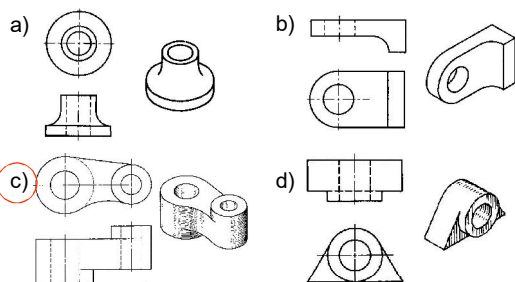
0 45 90 135 180

4. Which is in correct **third** angle projection ? (180 sec)



0 45 90 135 180

5. Which is a **wrong** 3rd angle orthographic views ? (180 sec)



0 45 90 135 180